American Computer Science League

2022-2023 ● Contest 2: Short Problems Solutions ● Intermediate Division

1. Prefix/Infix/Postfix Notation

6 3 / 2 2 ^ + 3 4 2 – * / = (6 3 /) (2 2 ^) + 3 (4 2 –) * /
= (6 / 3) (2 ^ 2) + 3 (4 – 2) * /
= (2 4 +) (3 2 *) / = (2 + 4) (3 * 2) /
= (6 6 /) = 6 / 6 = 1

B. 1

2. Prefix/Infix/Postfix Notation

4 6 + 7 5 – / 2 ^ 7 5 4 – * + = (4 6 +)(7 5 –) / 2 ^ 7 (5 4 –) * +
= ((+ 4 6)(– 7 5) /) 2 ^ (7 (– 5 4) *) +
= ((/ + 4 6 – 7 5) 2 ^) (* 7 – 5 4) +
= ((^ / + 4 6 – 7 5 2) (* 7 – 5 4) +)
= + ^ / + 4 6 – 7 5 2 * 7 – 5 4

E. +^/+46–752*7–54

3. Bit-String Flicking

(LSHIFT-2 (RCIRC-1 (NOT 10110))) = (LSHIFT-2 (RCIRC-1 01001))
= (LSHIFT-2 10100)
= 10000

B. 10000

4. Bit-String Flicking

Let X = abcde
LHS = (NOT 11011) AND X OR (RCIRC-1 01110)
= (NOT 11011) AND abcde OR (RCIRC-1 01110)
= (00100 AND abcde) OR 00111
= (00100 AND abcde) OR 00111
= 00c00 OR 00111
= 00111
LHS = RHS ⇒ 00111 = 00111 ⇒ a = *, b = *, c = *, d = *, e = *
Therefore x = *****

D. *****

5. LISP

(CDR (CAR (CDR '(b (c (a) b))))) = (CDR (CAR '((c (a) b))))
= (CDR '(c (a) b))
= ((a) b)

C. ((a) b)